

School Competition and Efficiency with Publicly Funded Catholic Schools[†]

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We study competition between two publicly funded school systems in Ontario, Canada: one that is open to all students, and one that is restricted to children of Catholic backgrounds. A simple model of competition between the competing systems predicts greater effort by school managers in areas with more Catholic families who are willing to switch systems. Consistent with this insight, we find significant effects of competitive pressure on test score gains between third and sixth grade. Our estimates imply that extending competition to all students would raise average test scores in sixth grade by 6 percent to 8 percent of a standard deviation. (JEL I21, I22, H75, Z12)

Rising costs and growing concerns over student achievement have led to renewed interest in the design of publicly funded education programs.¹ Building on the insights of Milton Friedman (1955), many analysts have suggested a voucher system that limits the monopoly power of local school districts. For a consumer-driven system to work, families must be willing and able to choose between schools to access higher quality education, and school administrators must be rewarded for attracting additional students. So far, the choice initiatives enacted in the United States have gone only part way toward satisfying these conditions.²

Alternatives to the local monopoly model of public school provision have a much longer history outside the United States.³ In this paper, we study the effects of competition among elementary schools in Ontario, Canada, which for over a

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¹See Eric A. Hanushek (2003) for an overview of trends in school spending and achievement in various countries.

²Two main initiatives to loosen the control of local school districts are charter schools and school vouchers. See Caroline M. Hoxby (2004), Kevin Booker et al. (2005), Robert Bifulco and Helen F. Ladd (2006), and Martin Carnoy et al. (2005) on the issue of charter schools, and see William G. Howell and Paul E. Peterson (2002), Alan B. Krueger and Pei Zhu (2003), and Ladd (2002) on the issue of vouchers.

³Damon Clark (2009) studies the efficiency of English high schools that opt out of local school district control, while Stephen Gibbons, Stephen Machin, and Olmo Silva (2008) analyze competition between public schools in England. Åsa Ahlin (2003) studies the effects of competition in Sweden. Chang-Tai Hsieh and Miguel Urquiola (2006) study the voucher system in Chile.

century has operated two parallel, publicly funded school systems: secular schools (known as *public schools*) that are open to all students, and *separate schools* that are open to children with Catholic backgrounds.⁴ The two systems are run independently, but follow a standardized curriculum and receive equal funding per student.

For non-Catholics, the Ontario system functions like a typical public system in the United States with a single monopoly provider. For the 40 percent of children with Catholic backgrounds, however, the system is effectively a voucher program with two competing suppliers. Although choice is limited to Catholics, the financial incentives to compete for Catholic students potentially impact the quality of schooling for *all students*. Our goal is to measure the effects of these incentives using standardized student test score gains between third and sixth grade.

We frame our analysis using a simple model of school choice and effort competition between public and separate schools. We assume that Catholic families choose between systems depending on a combination of tastes, proximity, and the quality of instruction. We also assume that administrators of the competing systems are rewarded for larger market shares. Under these conditions, equilibrium effort, and student achievement in both systems will be higher in areas with a larger share of Catholic families that are relatively uncommitted to a particular system, and are therefore more responsive to quality differentials between the two systems.

To identify the characteristics that are correlated with the willingness of Catholics to switch systems, we study the impact of school openings on enrollment trends at nearby schools. When Catholic parents are highly committed to one system or the other, the opening of a new school will have no effect on enrollment at nearby schools *in the competing system*. When they are relatively uncommitted, however, some families will switch systems. We find significant cross-system flows following new school openings, with a magnitude that is proportional to the local fraction of Catholics and is bigger in fast-growing areas, where families appear to have weaker ties to a particular system.

Building on these findings, we go on to test the prediction that test score gains between third and sixth grade are larger for students *in both systems* in more competitive markets (i.e., fast-growing neighborhoods with a higher fraction of Catholics). We specify an econometric model for individual test score gains that includes student, school, and neighborhood characteristics, as well as measures of potential competitive pressure based on the local fraction of Catholics and the neighborhood growth rate. The key identifying assumption is that variation in the local fraction of Catholics has no direct effect on average test score gains. We use two auxiliary datasets to investigate differences between Catholics and non-Catholics. We find that Catholic and Protestant parents and children have very similar characteristics, although both differ from other religious groups (in particular, people with no religious affiliation). Thus, our preferred specifications isolate the effect of variation in the local fraction of Catholics, holding constant the combined fraction of Catholics and Protestants in an area.

⁴ More precisely, we study the incentives for administrative effort and efficiency when students can choose between school systems. We follow the literature in referring to this as “school competition.”

We find statistically significant impacts of the market characteristics associated with greater potential competition on the growth rate of achievement between third and sixth grades. Both the fraction of Catholics and its interaction with a measure of population growth are associated with larger student test score gains. We verify that these effects extend to students in both systems, and that they are robust to controls for potential selection biases. Our estimates imply that expanding choice to all students in the province would have a modest effect on sixth grade test scores, raising achievement in sixth grade by 6–8 percent of a standard deviation.

I. Previous Research

Our research builds on several existing strands of research on public school performance. One set of studies addresses the effect of private school competition on public school achievement. Jim F. Couch, William F. Shughart II, and Al L. Williams (1993) related public-school test scores to the enrollment rate in private schools, and interpreted the positive correlation as a competition effect.⁵ Hoxby (1994) noted the potential endogeneity bias in this specification and used the local fraction of Catholics as an instrument for the private enrollment rate. Subsequent studies using a similar approach (e.g., Richard Arum 1996 and Christopher Jepsen 2003) have found weaker effects, and an extensive re-analysis by Jepsen (2002) concludes that the impact of private school competition in the United States is probably small. Our research design is similar and is intended to identify the *equilibrium effect* of potential competition. Importantly, however, we measure the effect on students *in both systems*, avoiding selectivity biases that would result if only the scores of students who remain in the public schools are considered.

A second group of studies, beginning with Melvin V. Borland and Roy M. Howsen (1992), examines Tiebout competition between public school districts in the same area.⁶ Although early studies tended to find small (or even a wrong signed) effects, Hoxby (2000) used the number of rivers and streams running through a metropolitan area as an instrument for the Herfindahl index of district enrollment shares, and obtained a positive competition effect. Jesse Rothstein (2007) re-analyzed these results and found smaller and generally insignificant effects. In a related study, Rothstein (2006) analyzed the effect of district fragmentation on sorting between districts and concluded that Tiebout competition effects are probably modest.

Direct evidence on voucher-based competition comes from a study of Chilean schools by Hsieh and Urquiola (2006). Using comparisons across municipalities they found no significant relationship between private school entry rates and district-wide gains in student achievement. Consistent with theoretical analyses by Dennis Epple and Richard E. Romano (1998) and Thomas J. Nechyba (2000), however,

⁵ Subsequent studies using the same approach (Craig M. Newmark 1995; William Sander 1999; Christopher R. Geller, David L. Sjoquist, and Mary Beth Walker 2006) have found generally insignificant effects. An exception is Thomas S. Dee (1998), who estimates a positive effect on graduation rates.

⁶ In an interesting district-level analysis, Daniel L. Millimet and Vasudha Rangaprasad (2007) test for strategic interactions between the input choices of nearby school districts in Illinois, and report positive and generally significant effects of nearby competitors' choices on a district's choices over pupil/teacher ratios, spending per pupil, and average school size.

they find that the introduction of vouchers led to an increase in the stratification of SES-groups across schools.

A third literature, starting with James S. Coleman, Thomas Hoffer, and Sally Kilgore (1982), compares the test scores of students at public and private schools. Arthur S. Goldberger and Glen G. Cain (1982) highlighted the potential selectivity biases in this comparison, and subsequent studies have used Catholic religion, distance to a Catholic school, and/or their interaction as instruments for private school choice (e.g., William N. Evans and Robert M. Schwab 1995, Derek Neal 1997, Jeffrey Grogger and Neal 2000, and David Figlio and Jens Ludwig 2000). Recently, Joseph G. Altonji, Todd E. Elder, and Christopher R. Taber (2005a) have shown that all three instruments are correlated with the graduation rates of Catholic students who attend public schools in eighth grade, calling into question the validity of the IV designs.

Like these studies, our identification strategy relies on comparisons between areas with different fractions of Catholic families, and could lead to biased inferences if the fraction of Catholics exerts an effect on test scores gains independent of any competition effect. To address the concerns raised by Altonji, Elder, and Taber's (2005a) findings, we focus on comparisons that vary the local fraction of Catholics, holding constant the combined fraction of Catholics and Protestants in an area. We show that in the US sample used by Altonji, Elder, and Taber (2005a), Catholics and Protestants have very similar test scores at eighth grade, and similar test score gains between eighth and tenth grade, although both groups have better outcomes than children with no religious affiliation. Likewise, Catholic and Protestant *parents* in Ontario have very similar earnings and education (though, again, both groups have better outcomes than parents with no affiliation). Both findings are consistent with our maintained assumption that variation in the fraction of Catholics, while holding constant the fraction of Protestants and Catholics, has no direct effect on average student achievement.

II. Institutional Detail and Conceptual Framework

A. Institutional Background

Ontario has had two publicly funded school systems since 1841. Originally both systems were financed by local property taxes, with ratepayers choosing which system received their property tax payments. An equalization program was introduced in the 1930s, and since 1985 the province has provided (roughly) equal per student funding for the two systems.⁷ Public schools are secular and are required to accept all students, whereas separate schools limit enrollment to children of Catholic backgrounds.⁸ The schools are administered by local school "boards" (equivalent to US school districts), with independent boards for the public system and the separate system. As of 2003, there were 31 public school boards in the province (with an average enrollment of 44,000 students) and 29 separate school boards (with an

⁷ Since 1998, the system has provided full provincial funding on a capitation basis.

⁸ Our understanding is that separate elementary schools strictly enforce this restriction.

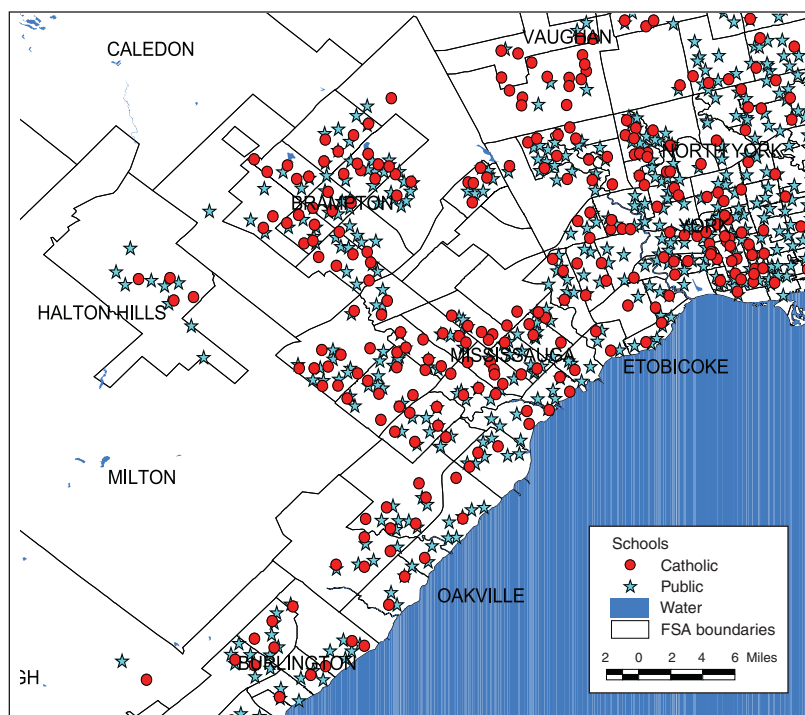


FIGURE 1. PUBLIC AND SEPARATE SCHOOLS, WEST OF TORONTO

average enrollment of 18,000).⁹ There were also 23 “school authorities” that operated schools in remote rural areas, and a handful of French-language school boards.¹⁰

Elementary teachers in the two systems are represented by separate unions (with board-specific collective agreements), but salaries and benefits are very similar across the province. Both systems also follow the same standardized curriculum, and administer the same set of standardized tests in mathematics, reading, and writing. School construction costs for both systems are financed by a provincial grant program that is designed to equalize the ratio of enrollment to capacity across boards.

Public and separate schools are readily accessible in all neighborhoods throughout the province. For example, Figure 1 shows the distribution of elementary schools in the cities and towns on the western edge of Toronto, superimposed over a graph of postal-code-based neighborhoods, known as Forward Sortation Areas, or FSAs. A typical FSA in this part of the province has eight to ten elementary schools, with a ratio of roughly two public schools per separate school. In both systems, children are assigned to schools using attendance zones. Thus, a given residential address falls into the attendance zone of *one* public elementary school and *one* separate

⁹ The boundaries of the public and separate boards typically coincide in more heavily populated areas.

¹⁰ About 4 percent of students attend the schools operated by French language boards. Another 5 percent attend private schools (which are mainly Protestant religious schools). We do not model the outcomes of students in private and French language schools, but we report models that include selection correction terms to control for the possible nonrandom selection into publicly funded English language schools.

elementary school.¹¹ Every Catholic family therefore has two basic choices for publicly funded education: their designated public school, or their designated separate school. Non-Catholic families have only one basic choice. In most areas, the boards also offer French immersion schools, though enrollment in these programs is small.¹² Finally, some boards allow students to attend schools other than their designated school, subject to space availability and other criteria. Participation in these “optional attendance” programs is very low.¹³

B. A Simple Model of Enrollment Demand, Managerial Effort, and School Quality

In this section, we present a simplified model of enrollment demand and effort determination in a stylized model with two competing schools: a public school open to all students, and a separate school open to Catholics. We show how changes in the fraction of Catholics, and in the degree to which these families are committed to one choice or the other, would be expected to affect the equilibrium efficiency of both schools. Although the model ignores other avenues of choice (in particular, the option of attending a private school), we believe it captures the salient features of the Ontario system. In Web Appendix 1, we extend the model to the situation where there are multiple schools operated by each board.

Consider an area with n_1 non-Catholic families and n_2 Catholic families, each with one school-age child. There are two available schools: a public school with quality Q_p and a separate school with quality Q_s . Catholic family i associates values U_{ip} and U_{is} to the choices, where

$$(1a) \quad U_{ip} = \alpha_{ip} + \beta Q_p - \gamma t_{ip} + \varepsilon_{ip}$$

$$(1b) \quad U_{is} = \alpha_{is} + \beta Q_s - \gamma t_{is} + \varepsilon_{is}.$$

Here, α_{ip} and α_{is} represent random taste components, t_{ip} and t_{is} represent travel costs, and $(\varepsilon_{ip}, \varepsilon_{is})$ are independently and identically distributed random shocks.¹⁴ Conditional on $(\alpha_{ip}, \alpha_{is}, Q_p, Q_s, t_{ip}, t_{is})$, the probability that family i selects the public school is $F[\delta_i + \beta\Delta Q - \gamma\Delta t_i]$, where F is the distribution function of the random variable, $v_i \equiv \varepsilon_{ip} - \varepsilon_{is}$, $\delta_i \equiv \alpha_{ip} - \alpha_{is}$ represents the family's relative taste for

¹¹ Similarly, every address is allocated to one public high school and one separate high school. The attendance zones of the competing systems typically do not coincide.

¹² French immersion programs are offered by the English language boards and are targeted at children whose parents are not French speaking. In a typical program students take some of their classes in French and others in English, depending on their grade. Fewer than 10 percent of public elementary students and under 5 percent of separate elementary students are enrolled in French immersion. We include these students in our test score analysis below.

¹³ For example, Halton District School Board (serving cities and towns on the western edge of Toronto) reported a total of 177 students in their optional enrollment program in the 2007–2008 school year, out of a total enrollment of about 35,000 elementary students. See Halton District School Board (2009).

¹⁴ A similar random utility formulation is widely used in models of demand for differentiated products. See Aviv Nevo (2000) for discussion and references, and Justine S. Hastings, Thomas J. Kane, and Douglas O. Staiger (2005) for an application to school choice.

public schools, ΔQ is the quality gap between the schools, and Δt_i is the difference in travel costs to the two schools.

Assume that the area is partitioned into a set of neighborhoods $k = 1, 2, \dots, K$, and that for all homes in neighborhood k the travel cost differential is Δt_k . The share of Catholic families in k who choose the public school is

$$(2) \quad s_k(\Delta Q, \Delta t_k) = \int F[\delta_i + \beta \Delta Q - \gamma \Delta t_k] h(\delta_i | k) d\delta_i,$$

where $h(\delta_i | k)$ is the density of relative tastes among Catholic families in the neighborhood.¹⁵ Letting n_{2k} represent the number of Catholic families in neighborhood k , the total fraction of Catholics who choose the public school is

$$s(\Delta Q) \equiv \sum_k \frac{n_{2k}}{n_2} \times s_k(\Delta Q, \Delta t_k).$$

Total public and separate school enrollments are

$$(3a) \quad E_p = n_1 + n_2 s(\Delta Q),$$

$$(3b) \quad E_s = n_2 [1 - s(\Delta Q)].$$

Using equation (2), the responsiveness of enrollment to the quality gap between schools is

$$(4) \quad \frac{\partial E_p}{\partial \Delta Q} = -\frac{\partial E_s}{\partial \Delta Q} = n_2 s'(\Delta Q),$$

where

$$s'(\Delta Q) = \sum_k \frac{n_{2k}}{n_2} \times \beta \int f[\delta_i + \beta \Delta Q - \gamma \Delta t_k] h(\delta_i | k) d\delta_i,$$

and $f[\cdot]$ is the density function associated with $F[\cdot]$.

Assume that school quality is an increasing concave function of the level of effort (e) exerted by school managers,

$$Q_\ell = q(e_\ell), \text{ for } \ell = \{s, p\},$$

and that preferences of each managerial team are represented by a linear function of effort and the *share* of local students attending their school,

$$U_\ell(E_\ell, e_\ell) = \theta \frac{E_\ell}{n} - e_\ell,$$

¹⁵ As an example, suppose δ_i is normally distributed with mean δ_k and variance σ^2 in neighborhood k . Then, $s_k = \int \sigma^{-1} F[\sigma z + \delta_k + \beta \Delta Q - \gamma \Delta t_k] \varphi(z) dz$, where $\varphi(z)$ is the standard normal density.

where $\theta > 0$ reflects the relative weight on market share.¹⁶ The first-order conditions for optimal effort are:

$$(5a) \quad \theta \left(\frac{n_2}{n} \right) s'(\Delta Q) q'(e_p) - 1 = 0$$

$$(5b) \quad \theta \left(\frac{n_2}{n} \right) s'(\Delta Q) q'(e_s) - 1 = 0.$$

In equilibrium, both teams supply the same level of effort e^* , with

$$(6) \quad q'(e^*) = \frac{1}{[\theta \left(\frac{n_2}{n} \right) s'(0)]}.$$

Since $q'(e)$ is decreasing in e , equilibrium effort is an increasing function of $\theta (n_2/n) s'(0)$, which depends on the strength of incentives faced by school managers (θ), the local fraction of Catholics (n_2/n), and the willingness of Catholic families to consider switching systems to access higher quality schools (i.e., the magnitude of s').

C. Assessing the Sensitivity of Enrollment Demand to Relative Quality

We do not directly observe responses of enrollment to variation in the relative quality of the public and separate schools. Under the assumptions of a standard characteristics-based choice model, however, it is possible to identify the features of the local population that lead to enhanced (or reduced) sensitivity to quality by studying how the same characteristics affect the reaction to changes in the proximity of different choices. The reason is that local demand responds to variation in quality and travel distance are both governed by heterogeneity in tastes for a religious versus secular education.¹⁷

To illustrate, suppose that a second separate school is opened with the same quality as the existing separate school. The effect on demand for enrollment at the *public* school is

$$\frac{\partial E_p}{\partial Open_s} = n_2 \sum_k \left(\frac{n_{2k}}{n_2} \right) \frac{\partial s_k(\Delta Q, \Delta t_k)}{\partial \Delta t_k} \times \frac{\partial \Delta t_k}{\partial Open_s},$$

where

$$\frac{\partial s_k(\Delta Q, \Delta t_k)}{\partial \Delta Q} = \beta \int f[\delta_i + \beta \Delta Q - \gamma \Delta t_k] h(\delta_i | k) d\delta_i$$

¹⁶ Equivalently, one could assume that managers are rewarded for the number of students they attract. Friedman's (1955) voucher proposal was predicated on the idea that managers receive a payoff proportional to the number of students they attract.

¹⁷ A similar idea is used by Randolph E. Bucklin, Gary J. Russell, and V. Srinivasan (1998) to show that the cross-price elasticities of market shares for competing brands is proportional to the probabilities of switching between brands.

and $\partial\Delta t_k/\partial Open_s$ represents the relative change in travel costs in neighborhood k to attend a public versus separate school.¹⁸ In neighborhoods that are closer to the new separate school than the old one, this expression is negative, leading some families to switch systems. From equation (2), however, the responses to changes in travel costs and quality are proportional:

$$(7) \quad \frac{\partial s_k(\Delta Q, \Delta t_k)}{\partial \Delta t_k} = -\frac{\gamma}{\beta} \frac{\partial s_k(\Delta Q, \Delta t_k)}{\partial \Delta Q}.$$

The enrollment loss when a nearby school is opened can be expressed as a weighted sum of the derivatives of the market shares in each neighborhood with respect to *quality*,

$$(8) \quad \frac{\partial E_p}{\partial Open_s} = -\frac{\gamma}{\beta} n_2 \sum_k \left(\frac{n_{2k}}{n_2} \right) \frac{\partial s_k(\Delta Q, \Delta t_k)}{\partial \Delta Q} \times \frac{\partial \Delta t_k}{\partial Open_s},$$

where the weight, $\partial\Delta t_k/\partial Open_s$, depends on the change in relative travel costs experienced by families in each neighborhood.

Assuming that the travel cost changes are a function of local geography, and do not co-vary with the distribution of tastes, we can use observed changes in enrollment following nearby openings by the competing system to infer the relative sensitivity of enrollment demand to quality. In particular, the same characteristics that are associated with bigger enrollment losses when a competing school is opened nearby would be expected to generate more elastic enrollment demand with respect to quality, leading to more intensive effort competition.

D. Extension and Implementation

Contrary to our simplified model, the public and separate systems operate multiple schools. Since both systems use attendance zones to assign addresses to specific schools, however, each family's choice problem is the same as in the two-school case. In particular, each Catholic family always faces two choices (their designated public and separate schools) while each non-Catholic family has only one choice (their designated public school). The effort-setting game is more complicated because each school manager competes with multiple managers in the opposing system (depending on the overlap of the attendance zones in the competing systems). Nevertheless, as we show in Appendix 1, the equilibrium has the same qualitative properties as in the two-school case. In particular, the incentives for effort depend on the local fraction of Catholic students, and on the degree of taste heterogeneity among Catholic families for a secular versus Catholic education. These same characteristics also affect the magnitude of the enrollment losses experienced by an existing school when the competing system opens a new school nearby. We use this

¹⁸ For simplicity, we are assuming that the changes in travel times $d\Delta t_k$ are small enough that the change in $\int F[\delta_i + \beta\Delta Q - \gamma\Delta t_k] h(\delta_i|k) d\delta_i$ can be approximated by $-\gamma d\Delta t_k \int f[\delta_i + \beta\Delta Q - \gamma\Delta t_k] h(\delta_i|k) d\delta_i = -\gamma d\Delta t_k \partial s_k / \partial \Delta t_k$.

TABLE 1—SUMMARY STATISTICS ON OPENING AND CLOSING SCHOOLS

	Total number of events (1)	Share of events 1998 or later (2)	Share of events with nearby nonrural school(s) (3)	Mean enrollment (schools with nearby nonrural schools) (4)
Public opening	252	0.587	0.631	304.3
Separate opening	169	0.604	0.633	273.5
Public closing	211	0.749	0.460	150.7
Separate closing	102	0.765	0.775	117.7

Notes: See text for definitions. Enrollment measure used in column 4 is average combined enrollment in grades 1–6 observed at opened or closed school during sample period.

insight to develop measures of the cross-system competitive pressure in different geographic areas.

III. Enrollment Responses to Nearby Openings and Closings

In this section, we use detailed enrollment data for public and separate elementary schools to study the effects of nearby openings and closings on *cross-system* enrollment shifts. Our goal is to identify the demographic characteristics of residents that are associated with more or less flexibility in the system preferences of Catholic families. We then use these results in Section IV to specify an empirical model of cross-system competition effects on test scores.

A. Identification of Openings, Closings, and Affected Schools

We obtained information on annual enrollments by grade at all publicly funded schools in Ontario for the period after 1990. We used these data to identify elementary schools that opened or closed from 1990 to 2004.¹⁹ (Details are presented in Web Appendix 2.) Table 1 summarizes the set of 421 elementary school openings and 314 closings that occurred over these 15 years.²⁰ As shown in the second column of the table, closings were particularly likely in the second half of the sample period, reflecting a consolidation of school boards that took place in 1998. Geographically, openings are concentrated in the growing cities and suburbs around Toronto, while closings were concentrated in outlying rural areas and in inner-city Toronto. We also matched opening and closing schools to census data tabulated at the FSA level to compare areas with openings and closings. Not surprisingly, openings tend to occur in areas with a high fraction of newly built houses, while closings are more likely in slow-growth areas.²¹

¹⁹ In brief, we define the opening year for a school as the first year with positive enrollment in grades 2–4, and the closing year as the last year with positive enrollment in these grades. For administrative purposes, schools are sometimes paired. We do not count these events as openings or closings.

²⁰ The sample includes all schools operated by English language school boards. During the 1990s, there were roughly 2,300 public elementary schools and 1,100 separate schools in operation in each year.

²¹ See Appendix Table 1. There is also some variation in income and family structure. Neighborhoods with only openings have the highest average income and lowest fraction of single parents, whereas those with only closings have the lowest family incomes and the most single parents.

Focusing on nonrural areas, we then identified nearby schools that were potentially affected by an opening or closing event. We began by identifying schools within a circle of radius equal to the average travel distance from home to school for students at local elementary schools (typically around 1–2 kilometers).²² We then used satellite images and maps to eliminate potentially affected schools that were separated from the newly opened or closed school by a major travel barrier (see Appendix 3). We have checked the sensitivity of our results to the inclusion of these “rejected” schools and find similar (though typically weaker) evidence of cross-system enrollment flows when they are included.²³ The mean distance from opened schools to affected schools is about 1.2 kilometers, while the mean distance from closed schools to affected schools is about 1.1 kilometers.

The third column in Table 1 shows the fractions of opening and closing events in nonrural areas for which we were able to identify at least one affected school. This ranges from 46 percent to 78 percent. Column 4 shows average total enrollment (for grades 1–6) at the opened and closed schools that have a nearby affected school. Newly opened schools have about 300 students (roughly 2 classes per grade), while the closing schools are smaller.

B. Enrollment Effects of Nearby Openings and Closings

We identified 945 nonrural elementary schools that were in operation for at least two years and were affected by a nearby opening or closing event between 1990 and 2004.²⁴ Using this sample, we estimate enrollment models of the form

$$(9) \quad \Delta E_{sat} = X_{sat}b + \sum_{j=1}^4 Event_{jst} \times \{D_s d_{j-pub} + (1 - D_s) d_{j-sep}\} \\ + \alpha_s + \omega_t + e_{sat},$$

where ΔE_{sat} is a measure of enrollment growth at affected school s in area a in year t ; X_{sat} is a vector of time-varying school and area characteristics; $Event_{jst}$ is a set of dummies for recent opening/closing events at nearby schools; D_s is an indicator equal to 1 if school s is public; α_s and ω_t are school and period fixed effects; and e_{sat} is an error term. We consider four types of events—public openings, separate openings, public closings, and separate closings—and allow separate coefficients (d_{j-pub} and d_{j-sep}) for the effect of each event type on nearby public and separate schools. Note that since the dependent variable is a growth rate, and the models include

²² We obtained information for one year on the postal codes of all students attending each elementary school in the province. We use the centroids of the postal codes for the schools and the homes to compute local travel distances.

²³ We also created a dataset that included only the schools near an opening or closing that are excluded from the main analysis sample. The estimated effects of openings and closings on these “excluded” schools are small and statistically insignificant.

²⁴ A cross-classification of affected schools by the number of “affecting events” (i.e., nearby openings and closings) is presented in Appendix Table 2. Two-thirds were affected by only one event, and another quarter was affected by exactly two events. Only about 10 percent were affected by 3 or more opening/closing events.

TABLE 2—SUMMARY STATISTICS ON ENROLLMENT GROWTH MEASURES

	Public schools (1)	Separate schools (2)
Enrollment in grade 1	49.1	43.9
(Standard deviation)	(24.3)	(21.3)
Number of school-year observations	7,746	5,332
Total enrollment in grades 1–6	284.8	262.4
(Standard deviation)	(117.4)	(109.1)
Number of school-year observations	7,746	5,332
Proportional change in grade 1 enrollment from previous year to current year ($\times 100$)	1.87	2.38
(Standard deviation)	(26.41)	(30.54)
Number of school-year observations	6,994	4,893
Proportional change in enrollment from grades 1–5 previous year to grades 2–6 current year ($\times 100$)	0.13	0.42
(Standard deviation)	(17.35)	(12.49)
Number of school-year observations	7,067	4,940

Note: Sample includes school-year observations for nonrural elementary schools affected by at least one opening or closing of nearby school over the sample period (1990–2004).

school fixed effects, these coefficients measure deviations of enrollment from a school-specific trend in the period following nearby opening or closing events.

Our primary dependent variable is the percentage change in enrollment from grades 1–5 in the previous year to grades 2–6 in the current year.²⁵ As a check, we also use the change in grade 1 enrollment from the previous year to the current year. This is somewhat noisier than the change in continuing enrollment, reflecting fluctuations in the size of the grade 1 entry cohort. To the extent that parents make a “once and for all” selection when their children first enter school, however, the growth in first grade enrollment may be a better indicator of sensitivity to changes in the presence of other local schools.

Table 2 presents summary statistics for these two enrollment growth measures. A typical public school in the sample has about 49 students per grade, while a typical separate school is a little smaller (44 students per grade). Between consecutive years grade 1 enrollment rises at an average rate of about 2 percent, with substantial variation across schools (the standard deviation is 26 percent for public schools and 31 percent for separate schools). Average enrollment growth from grades 1–5 to grades 2–6 is a little smaller (under 1 percent) and less variable.

The control variables in equation (9) are mean FSA characteristics interpolated from the 1991, 1996, 2001, and 2006 censuses.²⁶ These include the shares of the population aged 5–9 and 10–14 in the neighborhood, the share of recently constructed houses, and various demographic characteristics, including the fraction of

²⁵ Our school level database includes enrollment by grade for each school, as well as the enrollment of ungraded students in special education and other programs. We allocated these students uniformly across all grades offered by the school. For schools that do not offer all grades from 1–6, we modify the enrollment measure to reflect only those grades for which the school consistently has enrollment.

²⁶ In assigning schools to FSA's, we use 1996 boundaries. See Appendix 2 for more details. We link FSAs across census years using information provided by Statistics Canada. In cases where FSA-level data are unavailable in a year, we use values from the closest available year. Summary statistics for all variables are in Appendix Table 1.

TABLE 3—ENROLLMENT GROWTH MODELS

	Enrollment growth measure:						
	Change in 1st grade enrollment	Change in enrollment from grades 1–5 in year $(t - 1)$ to grades 2–6 in year (t)					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Own-system effects:</i>							
Effect of public opening on nearby public school	–7.8 (1.3)	–6.3 (0.7)	–2.9 (1.2)	–6.3 (0.7)	–6.3 (0.7)	–2.9 (1.2)	–2.9 (1.2)
Interacted with share new housing stock in FSA	—	—	–13.2 (4.1)	—	—	–12.8 (4.1)	–12.8 (4.1)
Effect of separate opening on nearby separate school	–9.7 (2.0)	–9.6 (1.3)	–9.0 (2.3)	–9.5 (1.3)	–9.5 (1.3)	–9.1 (2.3)	–9.1 (2.3)
Interacted with share new housing stock in FSA	—	—	–1.4 (5.9)	—	—	–1.0 (5.9)	–1.1 (5.9)
<i>Cross-system effects:</i>							
Effect of public opening on nearby separate school	–2.0 (1.3)	–1.1 (0.6)	1.1 (0.9)	0.1 (2.2)	—	—	—
Interacted with share new housing stock in FSA	—	—	–7.4 (2.5)	—	—	—	—
Interacted with share of Catholics in FSA	—	—	—	–3.2 (5.9)	–3.0 (1.7)	2.3 (2.5)	—
Interacted with share of Catholics in FSA × share new housing stock in FSA	—	—	—	—	—	–15.6 (6.8)	–11.1 (4.4)
Effect of separate opening on nearby public school	–3.3 (1.5)	–2.9 (0.9)	1.7 (2.0)	–3.0 (4.2)	—	—	—
Interacted with share new housing stock in FSA	—	—	–14.2 (7.3)	—	—	—	—
Interacted with share of Catholics in FSA	—	—	—	0.4 (12.0)	–7.7 (2.5)	7.5 (5.7)	—
Interacted with share of Catholics in FSA × share new housing stock in FSA	—	—	—	—	—	–42.9 (19.0)	–27.4 (9.2)
Observations	11,887	12,007	12,007	12,007	12,007	12,007	12,007
Number of schools	939	945	945	945	945	945	945

Notes: Standard errors are in parentheses. All models also include the following controls: fixed effects for schools and sample year; indicators for being affected by a nearby closing of a school in the same system or the competing system; a dummy for being paired with another school for administrative purposes; and the following characteristics of the local area (matched by FSA—see text): share of enrolled students in the FSA attending public French and private schools, total population in the FSA and shares of population ages 5–9 and 10–14; fraction of FSA residents who are Catholic; fraction who are immigrants; fractions of FSA residents of East Asian, South Asian, and Northern, Southern, and Eastern European ancestry; fraction of population with a university degree; fraction with no high school degree; fraction of single-headed families; fraction of families with one or two kids; and fraction of adults with home language other than English. British or French ancestry is treated as equivalent to “Canadian.” Eastern European ancestry groups include countries formerly affiliated with the USSR.

Catholics. In addition, we use our enrollment database to estimate the local fractions of children enrolled in public Francophone schools and private schools.

An issue for the specification of equation (9) is the duration of the effects of openings and closings on enrollment trends at nearby schools. We found that the cross-system effects emerge with some lag over a period of at least three years, whereas the effects on schools in the same system occur quickly (as might be expected if attendance zones are adjusted immediately after openings and closings). We therefore allow the opening and closing events to affect enrollment at affected schools in each of the three following years, with no impact before and after. Our findings are very similar if we allow four-or five-year impacts.

Table 3 shows the estimated effects of school openings on enrollment changes at nearby schools in the same and in the opposing system from a number of different variants of equation (9). All the models also include effects for nearby closings. As discussed in more detail below, however, the *cross-system* effects of closings are never large, and, in the interest of space, we report the closing effect coefficients in Web Appendix Table 3. Columns 1 and 2 of Table 3 present basic models that assume a constant effect of nearby openings and closings on the change in grade 1 enrollment (column 1) or the change in continuing enrollment between grades 1–5 and 2–6 (column 2). The estimates in column 1 suggest that nearby openings cause losses in first grade enrollment of 8 percent to 9 percent per year at nearby schools in the same system in each of the three years after the event (i.e., a cumulative impact of about 25 percent). More interesting from our perspective are the cross-system effects, which are also negative, but about one-third as large (i.e., losses of 2 percent to 3 percent per year). The cross-system effects on continuing-cohort enrollment changes are a little smaller, potentially reflecting the “stickiness” of system choice once students have started at a particular school.

Although not reported in the table, the estimated closing effects from the models in columns 1 and 2 show enrollment gains of 13–20 percent per year for the next 3 years following the closing of a nearby school in the same system. Unlike openings, however, closings seem to have no systematic cross-system effects. One simple explanation for the difference is that parents have a preference for new schools. In this case, we would expect to see larger enrollment losses following a nearby opening than gains following a nearby closing. To test this explanation we estimated the cross-system responses to openings, allowing different impacts at newer affected schools (those under five years old) and older schools. Contrary to our initial expectations, the estimates showed somewhat *larger* losses at relatively new schools.

An alternative explanation is that cross-system flows are driven by the decisions of recently arrived Catholic families who are weakly connected to existing schools. Since openings tend to occur in fast-growing areas with a higher fraction of new residents, while closings occur in slow-growing areas, the cross-system response to nearby openings will be relatively larger than the response to closings. This hypothesis also explains the tendency for bigger cross-system enrollment losses at newer schools, since these tend to be located in fast-growing areas.

To test this explanation, we fit a model that allows the effect of nearby openings to vary with a simple proxy for the newness of the neighborhood—the growth rate of the housing stock over the 1990s, as measured by the fraction of units in 2001 that were built after 1991.²⁷ The results are presented in column 3 of Table 3. The interaction coefficients show that nearby openings have larger (i.e., more negative) cross-system effects in newer neighborhoods.²⁸ Interestingly, the implied responses in slow-growth areas are essentially zero—comparable to the cross-system enrollment flows following school closings.

²⁷ We have also used an alternative based on local population growth. This yields qualitatively similar, but somewhat less precise, estimates for the models presented below.

²⁸ The mean of the new housing share variable is 0.17, with a range between 0.01 and 0.98, and a standard deviation of 0.18. The new housing share in neighborhoods with school openings is about five times bigger than in neighborhoods with school closings.

Since only Catholic students can switch between school systems, the cross-system enrollment losses from nearby openings should vary proportionally with the fraction of Catholics in the local area.²⁹ To test this, we include interactions of the cross-system opening indicators with the fraction of Catholics in the FSA in which the school is located. Results for four versions of this specification are reported in columns 4–7 of Table 3. The specification in column 4 allows the cross-system effect of a recent opening to depend on a dummy and an interaction with the local fraction of Catholics. Judging by the sampling errors, we do not have enough power to separately identify the two effects, so we fit the restricted model in column 5 that excludes the indicator, effectively *imposing* the assumption that cross-system enrollment responses are proportional to the fraction of Catholics in the area. The estimated coefficients suggest that in very high Catholic areas, the cross-system reactions to nearby openings are relatively large, roughly one-half the magnitude of the own-system responses.

The model in column 6 of Table 3 allows the cross effects to vary with both the local Catholic share and the growth rate of the local housing stock. Echoing the results in column 3, the triple interaction terms (in rows 8 and 12) are large in magnitude and statistically significant, whereas the two-way interactions of the Catholic share with the opening indicator (in rows 7 and 11) are small. Our final specification in column 7 therefore includes only the triple interaction $\text{Opening} \times \text{Catholic Share} \times \text{New Housing Share}$. This model fits as well as the preceding one, and suggests that schools in neighborhoods with more Catholic families and a larger fraction of new homes face a greater threat of enrollment losses to the competing system.

IV. Impacts of Competition on Student Achievement

In this section, we turn to the primary focus of our empirical analysis, which is to test whether potential competition between the competing school systems leads to bigger test score gains. Building on the insights of our theoretical model, and the results in the previous section, we develop indexes of competitive pressure in a local area that depend on the fraction of Catholics and the fraction of new homes in an area.

A. Modeling the Effect of Effort Competition on Student Achievement

Since 1998, students at publicly funded schools in Ontario have taken written standardized achievement tests in grades 3, 6, and 9. We use the grade 3 and grade 6 results to measure the “value-added” of elementary schools. Building on the existing literature, we assume that the test score of student i in grade $g \in \{3, 6\}$ who attends school s in area a depends on his or her observed characteristics X_{isa} , on school characteristics Z_{sa} (including an indicator for whether the school is public or separate),

²⁹ Strict proportionality will only hold if the distribution of preferences does not vary with the fraction of Catholics.

on the characteristics of the local area W_a , on a measure of competitive pressure in the area, I_a , and on unobserved ability and random factors e_{gisa} :

$$(10) \quad T_{gisa} = X_{gisa}b_{gx} + Z_{sa}b_{gz} + W_ab_{gw} + I_ab_{gl} + e_{gisa}.$$

Consistent with our theoretical model, competitive pressure in this specification is a *market characteristic* that affects the achievement of all students, not just those who attend a specific school. Note too that the coefficients in equation (10) are all grade-specific. Assuming that greater competitive pressure leads to a better learning environment, we expect that $b_{6l} > b_{3l}$.

Because of neighborhood sorting, unobserved student abilities are likely to be correlated with school and neighborhood characteristics, leading to biases in OLS estimation of (10). When longitudinal student data are available, the conventional solution is to estimate the model in first differences (e.g., Steven G. Rivkin, Hanushek, and John F. Kain 2005). Since we cannot link third and sixth grade test takers in most years of our database, we follow an alternative approach of including school $\times$ cohort fixed effects and entering the time-invariant school and neighborhood variables as *interactions* with an indicator for test takers in sixth grade ($Gr6_{ig}$):

$$(11) \quad T_{gisa} = X_{gisa}b_{gx} + Z_{sa}Gr6_{ig}b_z + W_aGr6_{ig}b_w + I_aGr6_{ig}b_l \\ + \xi_{coh(i),s} + e'_{gisa}.$$

In this specification, $\xi_{coh(i),s}$ represents a fixed effect for the cohort of students who were in third grade at school s at the same time as student i (or in sixth grade at the same school three years later), and e'_{gisa} represents the unexplained component of student i 's score in grade g . In the presence of cohort $\times$ school fixed effects, we can only identify the differential (or “value-added”) effects of the time-invariant variables. Thus, the coefficient on the competitive index is $b_l = b_{6l} - b_{3l}$, with similar expressions for b_z and b_w .

If all students remained at the same school between third and sixth grades, estimates based on this approach would be numerically identical to those from a model of test score gains between third and sixth grades. In the presence of student mobility, however, the two approaches will yield different estimates. For consistent estimation of the value-added coefficients using (11), any difference in average abilities between the leavers and joiners at a school must be uncorrelated with the index of competition—an assumption that may or may not be true.

Although longitudinal student identifiers are unavailable for most years of our sample, we obtained access to a file that allows us to link sixth grade test takers in 2007 with their third grade outcomes in 2004. This “validation sample” allows us to compare the test score gains of the entire cohort of students at a given school with the gains for the stayers only, and to test whether use of the cohort-based data leads to any bias in the estimated effect of our key independent variables. Full details are presented in Appendix 4. In brief, we compare the test score gains for all students

at a given school between third and sixth grades with the change for the roughly 70 percent of stayers. We then regress this gap on the same covariates included in our main value-added models. Reassuringly, we find that the estimated effects of our competition measures are insignificantly different from zero. In fact, the point estimates suggest that a full cohort analysis tends to *under-state* the effects of competition on the stayers only. We use the implied estimates of the bias effects to construct “corrected” estimates below.

A second key assumption is that our measures of competitive pressure are orthogonal to any unobserved determinants of achievement *growth*. Based on the analysis in Table 3, we use two measures of local competitive pressure. The first is just the fraction of Catholic families in an area. The second is the product of the local fraction of Catholics and the growth rate of the local area. Since we do not observe the religion of individual test takers, either of these choices poses a potential problem if Catholic children have systematically different test score gains than other students, or if the presence of more Catholics exerts an independent effect on the efficiency of schools. We study this issue in detail in Section IVC.

B. Test Score Data

We obtained individual test results for third and sixth grade students at all elementary schools in the province from 1998 to 2005, allowing us to track five cohorts of students, starting with children who were in third grade in the 1997–1998 school year and ending with those in third grade in 2001.³⁰ The sample includes roughly 65,000 public school students and 32,000 separate school students in each cohort. The test data file has a limited set of individual characteristics, including gender, whether a student is classified as “exceptional” (i.e., special needs) or “gifted” (i.e., advanced), whether he or she attended kindergarten, and whether he/she is enrolled in an English-as-a-Second-Language (ESL) or French immersion program. (We do not know whether a student is Catholic.) Means for these characteristics by grade and public versus separate school status are shown in the upper rows of Table 4. There are some small but statistically significant differences between students in the two systems. For example, separate school students are a little more likely to have attended kindergarten, are less likely to be classified as exceptional or gifted, and are less likely to be enrolled in ESL or French immersion.

We limit our analysis sample to children in school-cohort groups that have at least ten test takers in both third and sixth grades. We also compare the number of test takers in a cohort in third and sixth grades, and eliminate groups for which the ratio is greater than 140 percent or less than 71 percent.³¹ Our final sample includes 65–70 percent of all public school test takers and a higher fraction (85–96 percent)

³⁰ The tests are administered by the Education Quality and Accountability Office (EQAO). EQAO will not release test records for schools with fewer than 15 students enrolled in the grade of the test. Thus, our analysis does not include those schools with low enrollments.

³¹ We have estimated our main models using samples with different exclusion rules and find that the coefficients estimates are similar, though typically a little smaller in absolute value, when we retain school-cohorts with a wider range of variation in the number of test takers.

TABLE 4—SUMMARY STATISTICS FOR EQAO TEST TAKERS

	Public schools		Separate schools	
	Grade 3 (1)	Grade 6 (2)	Grade 3 (3)	Grade 6 (4)
<i>Test-taker characteristics for students with at least one test score:</i>				
Observations	323,508	340,259	164,502	172,409
Share female	0.49	0.49	0.49	0.49
Share ESL students	0.06	0.05	0.02	0.02
Share exceptional (special needs) students	0.04	0.04	0.03	0.03
Share attended kindergarten	0.86	0.73	0.89	0.75
Share French immersion students	0.09	0.06	0.04	0.04
Share gifted students	0.01	0.02	0.001	0.01
Share with scores for all three tests	0.88	0.94	0.91	0.95
<i>Reading tests</i>				
Observations	212,761	229,650	147,721	152,638
Average score (1–4 scale)	2.52	2.68	2.52	2.70
(standard deviation)	(0.76)	(0.74)	(0.75)	(0.73)
Share of students with missing score	0.13	0.09	0.10	0.07
<i>Mathematics tests</i>				
Observations	230,562	238,153	153,117	154,042
Average score (1–4 scale)	2.73	2.71	2.68	2.68
(standard deviation)	(0.75)	(0.80)	(0.73)	(0.79)
Share of students with missing score	0.09	0.08	0.08	0.06

Notes: Based on standardized tests administered in the period 1998–2005 to students in grades 3 and 6. Student observations are included for school-cohorts that have at least 10 test takers in grade 3 and 3 years later in grade 6, with the ratio of the number of test takers in grade 6 to the number in grade 3 between 0.71 and 1.40.

of all separate school test takers. Students in the sample are drawn from approximately 9,000 school-cohort groups who attended 2,000 different schools.

Summary statistics for the test outcomes of the analysis sample are presented in the bottom rows of Table 4.³² We show mean test scores by grade and public/separate school status for two of the three main test components: reading and mathematics.³³ The results are reported on a relatively coarse 4-point scale, with a mean of roughly 2.7 and a standard deviation of 0.6 to 0.8.³⁴ In contrast to the case in the United States, where students at Catholic schools tend to have higher scores than those at public schools, mean test scores are quite similar in the two systems. Presumably this reflects the much different selection process in the Ontario system, where Catholic schools are free and readily available in all neighborhoods. The similarity of scores between the separate schools (where 100 percent of children have a Catholic background) and the public schools (where only about 20 percent

³² Appendix Table 4 shows the mean test score outcomes for the overall samples of test takers and the fractions of students whose scores are missing. Many missing observations are attributable to exceptional (special needs) students, who are not required to take the test.

³³ The results for the writing scores are similar to those for reading. They are available from the authors.

³⁴ This limited scale poses a potential attenuation problem, although the fraction of students coded with the minimum score is less than 10 percent, while the fraction coded with the top score is less than 15 percent.

TABLE 5—RELATIONSHIP BETWEEN RELIGION AND EDUCATION/EARNINGS AMONG ONTARIO PARENTS

	Dependent variable = years education				Dependent variable = log weekly wages			
	Mothers		Fathers		Mothers		Fathers	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Catholic	0.13 (0.04)	−0.01 (0.04)	0.01 (0.04)	−0.06 (0.04)	0.04 (0.01)	0.02 (0.01)	0.03 (0.01)	0.01 (0.01)
Other religion (not Catholic or Protestant)	—	−0.20 (0.06)	—	0.08 (0.07)	—	−0.07 (0.02)	—	−0.08 (0.02)
No religion	—	−0.44 (0.05)	—	−0.32 (0.05)	—	−0.04 (0.02)	—	−0.06 (0.01)
Dummies for country of origin and metro area (CMA)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean of dependent variable [standard deviation]	14.16 [2.78]	14.16 [2.78]	14.20 [2.95]	14.20 [2.94]	6.21 [0.83]	6.21 [0.83]	6.75 [0.74]	6.75 [0.74]
Observations	31,744	31,744	27,988	27,988	22,652	22,652	23,467	23,467

Notes: Sample includes household heads and spouses age 24–64 in households with at least one child age 15 or less living in the province of Ontario in 2001 Canadian census. Dependent variable in columns 1–4 is estimated years of completed education. Dependent variable in columns 5–8 is log average weekly wage last year. For specifications in odd-numbered columns, omitted religious group is all non-Catholics (including those with no religion). For specifications in even-numbered columns, omitted religious group is Protestants. Models are fit by unweighted OLS. Standard errors are in parentheses.

are Catholic) suggests that, on average, Catholic children do not have characteristics that cause higher achievement.

C. Differences between Catholic and Non-Catholic Families

A critical assumption in our research design is that students in areas with a higher or lower fraction of Catholics have similar unobserved factors that affect achievement *growth*. While the similarity of the mean test scores in public and separate schools is consistent with this assumption, we cannot directly test it because we do not observe the religious affiliation of test-takers. Instead, we turn to two other data sources. The first is the 2001 Canadian census, which includes information for a large sample of Ontario residents on education, earnings, and religious affiliation.³⁵ Since parental education is a powerful predictor of test scores (e.g., Christopher Jencks and Meredith Phillips 1998), any differences in the education of Catholic versus non-Catholic parents would indicate a problem for our research design. Likewise, since wages are strongly affected by cognitive skills (Richard J. Murnane, John B. Willet, and Frank Levy 1995), comparisons of earnings potentially reveal differences in cognitive skills of parents that may be correlated with those of their children. Our second data source is the National Education Longitudinal Study of 1988 (NELS), which interviewed children in eighth grade in 1988 and followed them for the next two decades. Although NELS is US-based, it has the advantage of including test score information at different ages, as well as detailed information on family background and religion.

³⁵ The public use files of the 2001 Canadian census do not allow users to construct families. We classify as “parents” individuals between the ages of 24 and 62 who are either the head or spouse of the head of the household for a household with at least one child under the age of 16.

TABLE 6—MODELS FOR 8TH GRADE TEST SCORES AND TEST SCORE GAINS FROM 8TH TO 10TH GRADE, STUDENTS IN NELS-88

	All students				Students in public schools in 8th grade			
	8th grade tests		Test gain: 10th–8th		8th grade tests		Test gain: 10th–8th	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Catholic	120.5 (18.5)	33.9 (20.9)	37.4 (10.5)	17.0 (11.8)	102.0 (21.9)	14.4 (24.2)	35.9 (12.3)	14.7 (13.7)
Other religion (not Protestant or Catholic)	—	–39.0 (22.4)	—	–25.3 (12.7)	—	–22.3 (24.2)	—	–26.4 (13.6)
No religion	—	–57.2 (29.3)	—	–41.3 (16.6)	—	–41.4 (30.7)	—	–40.8 (17.3)
Religion not reported	—	–365.7 (23.8)	—	–68.9 (16.3)	—	–389.0 (25.2)	—	–66.5 (17.1)
Mean of dependent variable [standard deviation]	5,081.1 [988.5]		–33.3 [472.4]		4,973.6 [979.4]		–38.1 [470.1]	
Observations	13,315		12,037		11,078		9,970	

Notes: Dependent variable in columns 1–2 and 5–6 is composite test score (math and reading) in eighth grade. Dependent variable in columns 3–4 and 7–8 is change in composite test score from eighth to tenth grade. Models are fit by unweighted OLS and do not account for NELS sample design. For specifications in odd-numbered columns, omitted religious group is all other religions (including none and not reported). For specification in even-numbered columns, omitted religious group is any Protestant religion. All models include the following additional controls: dummies for gender, race/ethnicity, urban location, and census division; mother's and father's education (with allocated data for missing cases, and dummies for allocated status); dummy for living with mother and father at eighth grade, and dummies for SES quartile. Standard errors are in parentheses.

Table 5 presents comparisons of education and wages by religious affiliation for Ontario parents in the 2001 census. We present models in which the years of education or log weekly wages of a parent are regressed on dummies for religious affiliation, Census Metropolitan Area (CMA), the lowest level of geography identified in the public use file, and country of origin. The set of controls is relatively parsimonious, reflecting the controls in our test score models (see below). The odd-numbered columns of Table 5 present models that include only a dummy for Catholic religion. In these models, Catholics are compared to all other parents, including Protestants (about 35 percent of parents), those with no religious affiliation (about 15 percent of parents), and those with other affiliations (18 percent of parents). The specifications in the even-numbered columns include dummies for affiliates of other religions and nonaffiliates. In these models, Catholics are directly compared to Protestants. Looking across the columns, the estimates show that Catholic parents have a (modest) education and wage advantage relative to other parents as a whole, but are statistically indistinguishable from Protestant parents.

A very similar pattern holds with respect to test score outcomes of children in the NELS. Table 6 presents models for eighth grade test score outcomes of NELS students, and for their test score gains between eighth and tenth grade.³⁶ We present results for all students (in columns 1–4), and for the subset who were attending public schools in eighth grade (columns 5–8). The latter group is particularly interesting because few of them attend Catholic high schools (Altonji, Elder, and Taber 2005a, 2005b). Thus, differences in test score gains between eighth and tenth grade

³⁶ NELS also has test scores in twelfth grade, but a much larger fraction of students (25 percent) have missing test data.

TABLE 7—TEST SCORE GROWTH MODELS

Dependent variable: individual test score (4-point scale from 1 to 4, SD $\approx$ 0.75)	Reading				Mathematics			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Local competition measures:</i>								
Share of Catholics $\times$ grade 6	0.15 (0.05)	0.11 (0.06)	—	—	0.23 (0.07)	0.17 (0.08)	—	—
Share of Catholics $\times$ share new housing stock $\times$ grade 6	—	0.27 (0.16)	0.41 (0.14)	0.34 (0.15)		0.42 (0.23)	0.63 (0.20)	0.46 (0.21)
<i>Other controls:</i>								
Share with no religion $\times$ grade 6	—	—	—	−0.20 (0.11)	—	—	—	−0.53 (0.16)
Share with other religions $\times$ grade 6	—	—	—	0.06 (0.07)	—	—	—	0.21 (0.10)
Dummy = 1 if test taker in grade 6	0.005 (0.04)	0.02 (0.04)	0.07 (0.03)	0.10 (0.03)	−0.25 (0.05)	−0.23 (0.05)	−0.15 (0.04)	−0.07 (0.05)
Separate school $\times$ grade 6	0.02 (0.01)	0.02 (0.01)	0.03 (0.01)	0.03 (0.01)	0.02 (0.01)	0.02 (0.01)	0.03 (0.01)	0.02 (0.01)
Share of new housing stock $\times$ grade 6	−0.001 (0.02)	−0.10 (0.06)	−0.15 (0.06)	−0.14 (0.06)	0.06 (0.03)	−0.10 (0.09)	−0.18 (0.08)	−0.15 (0.09)
R^2	0.16	0.16	0.16	0.16	0.18	0.18	0.18	0.18
Observations	742,770				775,874			

Notes: Standard errors (clustered by school) are reported in parentheses. Dependent variable is student-level test score in reading (columns 1–4) or mathematics (columns 5–8) in grade 3 or grade 6. All models include individual-level controls, controls for the average characteristics of the students in each school-cohort, and FSA-level controls, as well as school-cohort fixed effects. See text for description of additional controls.

arguably measure the direct impact of a Catholic family background, rather than a combination of family background effects and Catholic schooling effects.

As shown by the estimates in columns 1 and 5, when Catholics are compared to all other children, they have significantly higher eighth grade test scores. Likewise, the models in columns 3 and 7 show faster test score gains between eighth and tenth grade. When the comparison group is narrowed to Protestants, however, the differences are much smaller and are uniformly insignificant. As with Ontario parents, Catholic and Protestant children in the NELS are quite similar, although both groups do better than children with other religions or no affiliation.³⁷ We interpret the NELS findings as strongly confirming the conclusions from Table 5. In particular, Catholic children in Ontario and in the United States appear to have similar family background characteristics and similar test scores to Protestants. Both groups have better family characteristics and better test scores than children affiliated with other religions or with no religions.³⁸

³⁷ We have fit a wide variety of alternative models to the NELS data, including models that are fit by weighted OLS, using cross-sectional or panel weights, and models for tenth grade scores that include eighth grade scores on the right-hand side. These models yield very similar results to the specifications reported in Table 6. Models for parental education in the NELS sample yield results that are quite similar to those in Table 5, though Catholic fathers have somewhat higher education than Protestant fathers. One important difference between Protestants in Canada and the United States is the higher fraction of “nonmainstream” Protestants in the United States. In the NELS data, children with Baptist and Pentecostal affiliation have lower test scores than mainstream Protestants, and their parents have lower education.

³⁸ See Evelyn L. Lehrer (2010) for a review of evidence from the United States, which generally concludes that religious affiliation has a positive effect on schooling outcomes.

D. Models for Test Score Outcomes

Table 7 presents estimation results for four alternative specifications of equation (11), fit separately to individual scores in reading and mathematics. In addition to the explanatory variables listed in the table, the models include a dummy for sixth grade test takers and eight student-level controls (gender, ESL status, French immersion status, gifted or exceptional status, a dummy for kindergarten attendance, and dummies for unknown gender and kindergarten attendance). These variables are all entered with grade-specific coefficients. We also include school-cohort means of the student variables, along with the mean fraction of the group with missing test scores and the fraction with a missing score who are coded as exceptional, all interacted with an indicator for grade 6. In addition, the models include the fraction of immigrants in the FSA, and the fractions in the FSA who report East Asian, South Asian, Northern European, Southern European, and Eastern European ethnicity, all interacted with grade 6 status.³⁹ The models include school $\times$ cohort fixed effects, with estimated standard errors clustered by school to allow for arbitrary correlation across the students from any one school.

Columns 1 and 5 present a basic model that uses the local fraction of Catholics as an indicator of competitive pressure. The estimated coefficients are statistically significant and suggest that sixth grade scores in reading are about 0.06 points higher, and scores in math are about 0.09 points higher, in a 60 percent Catholic neighborhood than in a 20 percent Catholic neighborhood, holding all else constant. Since the standard deviations of sixth grade scores across students are approximately 0.75, these represent effect sizes of 7 percent to 12 percent of a standard deviation.

The models in columns 2 and 6 include both the local fraction of Catholics and the fraction of Catholics interacted with the share of new housing in the neighborhood as indicators of competitive pressure. Consistent with the pattern in our enrollment growth models, the estimated interaction terms suggest that competitive pressure is much stronger in newer neighborhoods. The specifications in columns 3 and 7 exclude the Catholic share variable—a restriction that is not rejected at conventional significance levels for the reading models, but is rejected in the mathematics model. Again, all estimates of the competition effect are highly statistically significant. To interpret the magnitude of the estimated effects, consider the 0.41 coefficient estimate from the reading score model. Comparing an area with a 60 percent Catholic share and a 20 percent new housing share (close to the sample average) to an area with the same new housing share and only 20 percent Catholics, reading scores in sixth grade are increased by 0.03 points, or an effect size of 4 percent of a standard deviation. The implied effect in a relatively new neighborhood (35 percent new housing share) is 8 percent of a standard deviation. The predicted effects for mathematics are 54 percent larger.

Our investigation of differences between Catholics and non-Catholics suggested that Catholics are very similar to Protestants, but that both groups are advantaged

³⁹ As Canada is historically comprised of immigrants from the United Kingdom and France, we exclude ethnicities affiliated with these countries from our European ethnic measures. In addition, we include Southern European countries previously aligned with Russia under the Eastern European ethnic grouping.

TABLE 8—ALTERNATIVE SPECIFICATIONS FOR TEST SCORE GROWTH MODELS

Dependent variable: individual test score (4-point scale from 1 to 4, SD $\approx$ 0.75)	Reading			Mathematics		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Local competition measures:</i>						
Share of Catholics $\times$ share new housing stock $\times$ grade 6	0.34 (0.15)	0.32 (0.15)		0.46 (0.21)	0.43 (0.22)	
Share of Catholics $\times$ share new housing stock $\times$ grade 6 $\times$ public school			0.48 (0.18)			0.62 (0.27)
Share of Catholics $\times$ share new housing stock $\times$ grade 6 $\times$ separate school			0.33 (0.15)			0.44 (0.22)
<i>Selection correction term:</i>						
Local enrollment rate in English-speaking publicly funded schools		-0.06 (0.04)	-0.06 (0.04)		-0.08 (0.05)	-0.08 (0.05)
Controls for shares of other religious groups	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.16	0.16	0.16	0.18	0.18	0.18
Observations		742,770			775,874	

Notes: Standard errors (clustered by school) are reported in parentheses. See notes to Table 7. All models include individual-level controls, controls for the average characteristics of the students in each school-cohort, and FSA-level controls, as well as school-cohort fixed effects.

relative to families with other religious affiliations, or no affiliation. In view of this finding, we added two additional controls to the models in columns 4 and 8: the fraction of people in the FSA who express no religious affiliation, and the fraction affiliated with religions other than Protestant or Catholic (i.e., Judaism, Islam, Hinduism, etc.). In these models, the effect of local competition is identified by variation in the fraction of Catholics relative to Protestants, *holding constant the fractions with other religion and no religion*.

The addition of these two controls attenuates the coefficient of our competition indicator slightly in the models in reading, and somewhat more for mathematics, though the coefficients remain significant at conventional levels. Consistent with the models for NELS test outcomes, a higher local fraction of people who report no religious affiliation is associated with slower test score gains, particularly in mathematics. By comparison the effects of the share of families with religious affiliations other than Protestant or Catholic are slightly positive, but not statistically significant. Interpreting the magnitudes of the competition effects, a rise of 40 percentage points in the fraction of children with choice between the systems, is associated with sixth grade test outcomes that are 3 percent to 5 percent of a standard deviation higher in a neighborhood with an average share of new houses.

E. Specification Checks

One potential concern with the specifications in Table 7 is that they ignore potential sample selection biases arising from the decision of parents to opt out of English language public schools and choose either private schools or publicly funded French language schools. To address this we follow Reuben Gronau (1974) and augmented our models with a control function representing the fraction of elementary students in the local area (FSA) who attend English language public schools (interacted with

the dummy for grade 6).⁴⁰ Estimation results for a specification using the interaction between the fraction Catholic and the fraction of new homes in an area as the index of competition are presented in Table 8. For reference, columns 1 and 4 reproduce the baseline specifications from columns 4 and 8 of Table 7, while the corresponding models with the selection term are presented in columns 2 and 5. The estimated selection effects are all negative (but insignificant), suggesting that students who opt out of the English language public schools have lower test score gains than average. Since the opt-out rate is slightly higher in more competitive areas, there is a slight upward selection bias (~ 0.02) in the uncorrected estimates of the competition effects. On balance, however, we conclude that selection effects are relatively small and can be safely ignored.

A second specification issue is the assumption, implicit in the models in Table 7, that local competitive pressure has the same effect on test score gains of students at public and separate schools. While this is predicted by the simple symmetric equilibrium of our model, in principle the effects of local competitive pressure could be different in the two systems. To test this possibility, we report in columns 3 and 6 of Table 8 the estimates from models that allow for different effects of the index of local competition at public and separate schools. Interestingly, these models all show a larger effect of competition on gains in the *public* schools, though the differences are never statistically significant.

A final issue is the potential bias caused by our use of test score gains for synthetic cohorts of students at a school, rather than for stayers only. In Appendix 4, we evaluate the magnitude of this bias by estimating models similar to the ones in Tables 7 and 8, but using as a dependent variable the *gap* in the estimates of value-added at a given school based on the full cohort of students and the stayers. We can use the results from these models to construct “bias-corrected” estimates of the effect of competition on the test score gains of the stayers only. For example, using the basic specifications in columns 4 and 8 of Table 7, we obtain the following results:

	Table 7 coefficient	Estimated bias (Web Appendix 4 Table B)	Bias-corrected coefficient
Reading	0.34 (0.15)	−0.08 (0.09)	0.42 (0.24)
Mathematics	0.46 (0.21)	−0.14 (0.09)	0.50 (0.23)

Because there is a negative relationship between the bias in the full cohort estimate of value-added and the measure of competitive pressure, the bias-corrected competition effects are all larger in magnitude than the coefficients in Table 7,

⁴⁰ If test score gains were normally distributed, and the decision to attend English language public schools was based on a single index choice model with a normally distributed random term, then the selection bias in the observed test score gains for students in an area where a fraction p of students are enrolled in the English language public schools would be proportional to the function $\lambda(p) = \varphi(\Phi^{-1}(p))/p$, where $\Phi^{-1}(\cdot)$ is the inverse normal distribution function and $\varphi(\cdot)$ is the standard normal density. In our sample $p > 0.85$, and in this range, $\lambda(p)$ is essentially linear in p . We therefore use the estimate of p as the control function for selectivity biases.

though less precisely estimated. We conclude that the estimates in Table 7 are, if anything, slightly conservative.

VI. Summary and Conclusions

Can a reduction in the monopoly power of local school districts improve the efficiency of publicly funded education? We answer this question by studying the effects of school competition in Ontario, Canada, which operates two publicly funded school systems: public schools that are open to all students, and separate schools that are limited to children with Catholic backgrounds. The fraction of families who can exercise choice between the competing systems varies widely across the province, providing the basis for our research design.

Our estimation results reveal significant, but relatively modest, effects of enhanced competition on the test score gains of students. Comparing markets where only 20 percent of children have choice to markets where 60 percent can choose between systems, we estimate that the gain in test scores between third and sixth grades is increased by 0.03–0.05 of a standard deviation. The impacts are bigger in rapidly growing areas, where competition between schools for newly arriving students appears to be most intense. Importantly, we also find that potential competition raises test scores in both the public and separate systems, with somewhat larger impacts for public school students.

It is worth noting two limitations of our analysis. First, we can only measure test score gains over three years, or one-quarter of the time that most students spend in school. If similar effects were present at all stages of elementary and secondary schooling the benefits of competition would be commensurately greater. Second, it is possible that in more competitive markets, teachers and principals spend more time and effort preparing for standardized tests, and less on other aspects of learning. If “test skills” have limited intellectual value, the effort devoted to competing over test outcomes is socially wasteful, and the higher test score gains observed in more competitive markets may be counter-productive.

Our findings have at least two implications for the design of alternative public education systems. First, we have shown that a significant fraction of families are willing to move between publicly funded schools to access a combination of higher quality or more convenient schools. This willingness to move provides the basis for cross-system competition that can lead, at least in principle, to improved efficiency of publicly funded schools. Second, our results underscore the critical importance of research on the links between parental choice decisions and the incentives faced by competing school systems.

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